Example 3: Solve $(1+x)y'' + y' - y = 0$ around $x = 0$.

- Since the leading coefficient satisfies $1 + 0 = 1 \neq 0$, the point $x = 0$ is an ordinary point.
- Let $y = \sum_{n=0}^{\infty} a_n x^n$.
- Using this series with $y' = \sum_{n=0}^{\infty}(n+1)a_{n+1}x^n$ and $y'' = \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n$ in $y'' + xy'' + y' - y = 0$, we obtain the coefficient of x^n is

$$(n+2)(n+1)a_{n+2} + n(n+1)a_{n+1} + (n+1)a_{n+1} - a_n.$$

Thus

$$(n+2)(n+1)a_{n+2} + (n+1)^2 a_{n+1} - a_n = 0,$$

and the recurrence relation is

$$\boxed{a_{n+2} = \frac{a_n - (n+1)^2 a_{n+1}}{(n+2)(n+1)}}.$$

- The first two coefficients a_0 and a_1 are arbitrary.



Example 3: Solution

- The subsequent coefficients are

$$a_2 = \frac{a_0 - a_1}{2},$$

$$a_3 = \frac{a_1 - 4a_2}{6} = \frac{3a_1 - 2a_0}{6},$$

and

$$a_4 = \frac{a_2 - 9a_3}{12}.$$

- Therefore the series solution is

$$\boxed{y = a_0 + a_1 x + \frac{a_0 - a_1}{2} x^2 + \frac{3a_1 - 2a_0}{6} x^3 + a_4 x^4 + \cdots,}$$

where all higher coefficients are obtained from the recurrence relation.

- Hence the method gives two linearly independent solutions by taking $(a_0, a_1) = (1, 0)$ and $(0, 1)$.



Example 4: Solve $y'' + xy' + y = 0$ around $x = 0$.

- Let $y = \sum_{n=0}^{\infty} a_n x^n$ be the solution.
- Then, $y' = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$ and $y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$.
- We can write $xy' = \sum_{n=1}^{\infty} n a_n x^n$.
- Substitution gives $\sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} + n a_n + a_n\right] x^n = 0$.
- Therefore, $(n+2)(n+1) a_{n+2} + (n+1) a_n = 0$, or

$$\boxed{a_{n+2} = -\frac{a_n}{n+2}}.$$

- For the even coefficients,

$$a_2 = -\frac{a_0}{2}, \qquad a_4 = \frac{a_0}{8}, \qquad a_6 = -\frac{a_0}{48}, \ldots$$



Example 4: Solution

- Thus,

$$a_{2k} = \frac{(-1)^k a_0}{2^k k!}.$$

- For the odd coefficients,

$$a_3 = -\frac{a_1}{3}, \qquad a_5 = \frac{a_1}{15}, \qquad a_7 = -\frac{a_1}{105}, \cdots$$

and, more generally,

$$a_{2k+1} = \frac{(-1)^k a_1}{1 \cdot 3 \cdot 5 \cdots (2k+1)}.$$

- Hence

$$\boxed{\begin{aligned} y = C_1 &\left(1 - \frac{x^2}{2} + \frac{x^4}{8} - \frac{x^6}{48} + \cdots\right) \\ &+ C_2 \left(x - \frac{x^3}{3} + \frac{x^5}{15} - \frac{x^7}{105} + \cdots\right). \end{aligned}}$$



Example 5: Nonhomogeneous Equation with Variable Coefficients

- Solve $y'' + xy' + y = x$ by the power-series method about $x = 0$.
- By Example 4, we have already got y_c by taking the corresponding homogeneous DE.
- If we take y_p as the same series, then LHS is $\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)a_n] x^n$.
- Since the right-hand side is x, its power-series coefficients are

$$b_0 = 0, \qquad b_1 = 1, \qquad b_n = 0 \quad (n \geq 2).$$

- Consequently,

$$(n+2)(n+1)a_{n+2} + (n+1)a_n = b_n,$$

so that

$$\boxed{a_{n+2} = \frac{b_n - (n+1)a_n}{(n+2)(n+1)}.}$$



Example 5: Solution

- For $n = 0$,
$$2a_2 + a_0 = 0, \qquad a_2 = -\frac{a_0}{2}.$$
- For $n = 1$,
$$6a_3 + 2a_1 = 1,$$
and hence
$$a_3 = \frac{1}{6} - \frac{a_1}{3}.$$
- For $n \geq 2$,
$$a_{n+2} = -\frac{a_n}{n+2}.$$
- Taking $a_0 = a_1 = 0$ gives a particular solution beginning with
$$y_p = \frac{x^3}{6} - \frac{x^5}{30} + \frac{x^7}{210} - \frac{x^9}{1890} + \cdots.$$
- Therefore, the general solution is $y = y_c + y_p$.



Example 6: Legendre's Equation (Setup)

- Legendre's equation: $(1-x^2)y'' - 2xy' + n(n+1)y = 0$.
- Here $P(x) = \dfrac{-2x}{1-x^2}$, $Q(x) = \dfrac{n(n+1)}{1-x^2}$.
- Both analytic at $x = 0$, so $x = 0$ is an ordinary point.
- Singular points occur at $x = \pm 1$.
- Assume $y = \sum_{k=0}^{\infty} a_k x^k$ and substitute into the equation
- After shifting indices, the recurrence relation obtained is.

$$a_{k+2} = \frac{k(k+1) - n(n+1)}{(k+1)(k+2)}\, a_k.$$



Example 6: Legendre's Equation (Solution)

- The recurrence $a_{k+2} = \dfrac{k(k+1) - n(n+1)}{(k+1)(k+2)} a_k$ links alternate coefficients.
- Choosing $a_0 = 1, a_1 = 0$ gives an even series solution $y_1(x)$.
- Choosing $a_0 = 0, a_1 = 1$ gives an odd series solution $y_2(x)$.
- General solution: $y = a_0 y_1(x) + a_1 y_2(x)$.
- If n is a non-negative integer, one series terminates, and gives the **Legendre polynomial** $P_n(x)$ as a solution.
- Series converges for $|x| < 1$. (RC =1).
- Write the expressions for $P_n(x)$ - HOMEWORK.



Example 7: Hermite's Equation (Setup)

- Hermite's equation: $y'' - 2xy' + 2ny = 0$.
- Here $P(x) = -2x$ and $Q(x) = 2n$, both analytic everywhere.
- So $x = 0$ is an ordinary point, in fact the only type of point.
- Assume $y = \sum_{k=0}^{\infty} a_k x^k$.
- Substituting and shifting indices gives coefficient equation

$$(k+2)(k+1)a_{k+2} - 2ka_k + 2na_k = 0.$$

- Recurrence relation: $a_{k+2} = \dfrac{2(k-n)}{(k+1)(k+2)} a_k$.



Example 7: Hermite's Equation (Solution)

- Using $a_{k+2} = \dfrac{2(k-n)}{(k+1)(k+2)} a_k$ with a_0, a_1 arbitrary.
- Even series (from a_0) and odd series (from a_1) form two solutions.
- General solution: $y = a_0 y_1(x) + a_1 y_2(x)$.
- When n is a non-negative integer, one series terminates into a polynomial.
- These give the **Hermite polynomials** $H_n(x)$.
- Write the expressions for $H_n(x)$ - HOMEWORK.
- Series converges for all $x \in \mathbb{R}$.



Example 8: Airy's Equation (Setup)

- Airy's equation: $y'' - xy = 0$.
- Here $P(x) = 0$ and $Q(x) = -x$, both analytic everywhere.
- So $x = 0$ is an ordinary point.
- Assume $y = \sum_{n=0}^{\infty} a_n x^n$.
- Then $y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$ and $xy = \sum_{n=0}^{\infty} a_n x^{n+1}$.
- Shifting index in the second sum and comparing powers of x gives

$$2a_2 = 0, \qquad a_{n+2} = \frac{a_{n-1}}{(n+2)(n+1)} \quad (n \geq 1).$$



Example 8: Airy's Equation (Solution)

- From $2a_2 = 0$ we get $a_2 = 0$, and the relation $a_{n+2} = \dfrac{a_{n-1}}{(n+1)(n+2)}$.
- Coefficients split into three chains based on a_0, a_1, and multiples of $a_2 = 0$.
- This gives two independent series:

$$y_1(x) = a_0 \left(1 + \frac{x^3}{6} + \frac{x^6}{180} + \cdots\right)$$

$$y_2(x) = a_1 \left(x + \frac{x^4}{12} + \frac{x^7}{504} + \cdots\right)$$

- General solution: $y = a_0 y_1(x) + a_1 y_2(x)$.
- These combinations define the **Airy functions** $\mathrm{Ai}(x)$, $\mathrm{Bi}(x)$.
- Series converges for all $x \in \mathbb{R}$ since there is no singular point.

